



Integration of NDIR Sensing, GPS and Multi-Gyroscope Modules for Real-Time In-Vehicle Alcohol Detection and Driver Safety Enhancement

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ABSTRACT: Globally, alcohol-impaired driving is the leading preventable cause of road traffic fatalities, contributing to nearly 22% of such deaths. Conventional alcohol detection systems often face limitations, including high cost, slow response times, and bulky integration, which reduce their practicality in vehicles. This study presents the development of a multi-sensor embedded system that integrates a non-dispersive infrared (NDIR) sensor, a global positioning system (GPS) module, and multi-gyroscope steering wheel monitoring to enable real-time alcohol detection and driver safety enhancement. The NDIR sensor measures ethanol concentrations in exhaled breath with an accuracy above 90%, while the GPS module provides location tracking with 95% precision to facilitate rapid emergency response. The gyroscope sensors detect impaired steering patterns and erratic maneuvers with over 95% accuracy, allowing early identification of unsafe driving behavior. A microcontroller-based server collects sensor data, generates automated health and safety reports, and transmits alerts to law enforcement and family members in cases of intoxication or dangerous actions. The prototype demonstrated an average response time of three seconds, ensuring rapid intervention capability. By combining advanced sensing technologies into a compact and cost-effective design, this system offers a scalable approach to reducing alcohol-related crashes and improving global road safety.

1. INTRODUCTION

Road traffic accidents remain one of the leading causes of preventable mortality worldwide. According to the World Health Organization (WHO), more than 1.25 million individuals lose their lives annually in road traffic crashes, with alcohol consumption recognized as a major contributing factor. The International Traffic Safety Data and Analysis Group (IRTAD) estimates that approximately 22% of traffic-related fatalities involve alcohol-impaired drivers, translating to nearly 273,000 deaths each year. Beyond the human cost, alcohol-related crashes also impose a substantial economic burden, accounting for an estimated 2.6% of the global Gross Domestic Product (GDP), equivalent to over 340 billion USD in 2019. These figures underscore the urgent need for more effective technologies to detect and mitigate alcohol-impaired driving. Although this is an intuitive process for adults, the pioneering scientist Jean Piaget's experiments included the observation that children are limited to recognizing that an object still exists when it is out of sight at younger ages. Hence, for a long time, such a topic has fascinated many cognitive-developmental psychologists.

Traditional alcohol detection systems, such as ignition interlock devices and dashboard-mounted breathalyzers, have been developed to reduce impaired driving. While these systems have shown promise, they are often characterized by key limitations: high cost, delayed response times, bulky integration into vehicles, and in some cases, reduced measurement accuracy. Moreover, many of these systems only activate after alcohol consumption has already impaired the driver, thereby limiting their preventive impact. Such drawbacks highlight the necessity for a compact, efficient, and cost-effective solution that can be seamlessly integrated into modern vehicles. Competence-performance disparity, on the other hand, refers to the difference between the children's actual competence, i.e., their knowledge or understanding, and their actual performance, i.e., the demonstrations they showed in their actions. This disparity may be a result of the task demands in the experiments, where these demands couldn't accurately express the infant's competence through their performance (Wood & Power, 1987).

Recent advances in embedded systems and sensor technologies have created opportunities to overcome these challenges. Non-dispersive infrared (NDIR) sensors have emerged as a reliable method for detecting ethanol in exhaled breath due to their high specificity and accuracy. Similarly, global positioning system (GPS) modules can provide precise location tracking to ensure timely emergency response, while gyroscope sensors enable the detection of abnormal steering behavior associated with driver impairment. The integration of these technologies within an embedded system offers the potential for a holistic approach to alcohol detection and real-time driver safety monitoring. As evident in Table 1, Piaget claimed that children, at some point, don't show any response to the object when hidden, which means that since those infants don't see the object, this object doesn't exist anymore.

Smart In-Vehicle Detector

This study presents the design and development of a multi-sensor embedded system that integrates an NDIR alcohol sensor, GPS module, and multi-gyroscope steering wheel monitoring unit. The system is controlled by a microcontroller-based server, which processes sensor data, generates automated health and safety reports, and transmits alerts to law enforcement and family members when intoxication or unsafe driving behavior is detected. By combining these technologies into a compact, low-cost platform, this research aims to provide an efficient, scalable solution to reduce alcohol-related accidents and enhance road safety globally.

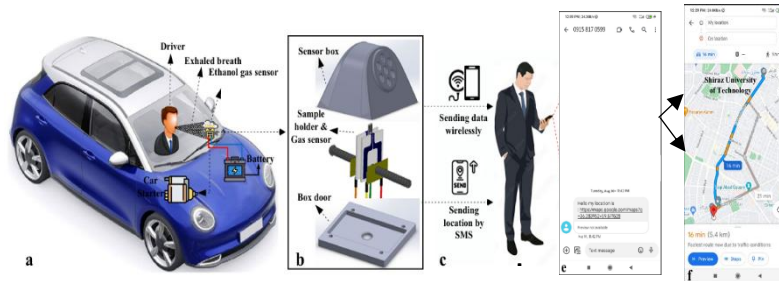


Figure 1. Summary of the project's methodology and system walkthrough.

2. LITERATURE REVIEW

Several technologies have been developed in recent decades to address the problem of alcohol-impaired driving. One widely explored approach is the breath-analyzed ignition interlock system, which prevents a vehicle from starting if alcohol is detected in the driver's breath. A study presented at the 2018 International Conference for Convergence in Technology (I2CT) demonstrated that this low-cost method effectively reduces alcohol-related accidents by requiring drivers to complete a breath test before ignition. However, its practicality is limited by slower response times, lower efficiency compared to advanced sensor-based systems, and potential inconvenience for drivers, particularly in emergency situations.

An alternative solution was proposed by Hitachi Ltd. in 2011, which involved the integration of a steering-wheel-mounted breath alcohol detection system. This approach relied on multiple modern sensors capable of analyzing alcohol levels from exhaled breath without requiring direct user input. While this system offered faster response times and improved usability compared to ignition interlocks, its drawbacks included significant cost and large space requirements within the vehicle's dashboard, making it less suitable for widespread adoption in consumer vehicles.

More recent work has explored passive in-vehicle alcohol detection using advanced sensor fusion techniques. For example, Ljungblad et al. (2017) demonstrated a system capable of analyzing lung-exhaled gases using non-dispersive infrared (NDIR) technology, achieving high efficiency and reduced space requirements. Although this approach outperformed earlier models in terms of accuracy and compactness, it remains among the most expensive solutions in the field. High costs of installation, testing, and sensor maintenance have limited its commercial feasibility.

Taken together, these prior studies highlight both the progress and the limitations of existing alcohol detection technologies. Low-cost systems often suffer from reduced accuracy or response time, while advanced sensor fusion methods provide high efficiency but at prohibitive costs. Furthermore, many existing solutions focus solely on breath analysis, without considering driver behavior monitoring or emergency communication systems. These gaps suggest the need for an integrated solution that combines affordability, high accuracy, compact design, and additional safety features.

The present study addresses this gap by developing a multi-sensor embedded system that integrates NDIR-based alcohol detection with GPS-enabled emergency response and multi-gyroscope driver behavior monitoring. By combining these complementary technologies into a single compact platform, the system aims to provide a practical and scalable solution for real-time alcohol detection and enhanced road safety.

3. METHODS

The proposed system was designed as a multi-sensor embedded platform capable of detecting alcohol in exhaled breath, monitoring driver behavior, and transmitting emergency alerts in real time. The methodology involved the integration of three core sensing units—an NDIR alcohol sensor, a GPS module, and multiple gyroscope sensors—controlled by a microcontroller-based server, as illustrated in Figure 1 (system architecture). The following subsections describe the implementation of each component and the system workflow.

The detection of ethanol concentration in exhaled breath was achieved using a non-dispersive infrared (NDIR) sensor, which operates on the principle of infrared light absorption. Ethanol molecules absorb light at specific wavelengths, and the sensor quantifies the concentration of ethanol in the sample by measuring the reduction in transmitted IR intensity.

The NDIR sensor used in this study achieved a detection accuracy exceeding 90% in controlled testing conditions. To evaluate performance, the sensor was calibrated using gas pulses of known ethanol concentrations, ranging from 50 ppm to 300 ppm, which correspond to blood alcohol concentration (BAC) values typically associated with impaired driving. A response time of approximately three seconds was observed, ensuring rapid analysis suitable for in-vehicle applications.

To provide real-time geolocation, a GPS module was integrated into the system. This module was capable of delivering longitude and latitude readings with an accuracy of approximately 95%, ensuring precise location tracking in both static and dynamic testing environments.

In the event of alcohol detection above a predefined threshold, the GPS module automatically transmits the driver's location as part of the safety alert protocol. This feature ensures timely communication with emergency responders and family members, significantly increasing the chances of intervention before accidents occur.

Impaired or unsafe driving behavior was monitored using four gyroscope sensors embedded in the steering wheel. These sensors detected angular velocity and sudden deviations in steering patterns within a sensitivity range of ± 1 to ± 10 degrees per second.

Experimental calibration demonstrated that abnormal steering maneuvers—characterized by erratic or delayed corrective actions—were detected with an accuracy exceeding 95%. The integration of multiple gyroscopes ensured redundancy and increased robustness against false positives caused by road conditions or minor driver adjustments.

A microcontroller-based server was employed to integrate sensor outputs and execute the decision-making algorithms. The server processed alcohol concentration data from the NDIR sensor, cross-referenced it with steering stability measurements from the gyroscopes, and verified the driver's location through GPS.

Upon detection of intoxication or unsafe driving patterns, the system automatically generated a health and safety report that included:

- Detected ethanol concentration,
- Abnormal steering behavior indicators, and
- GPS coordinates of the vehicle.

This report was transmitted via a GSM-enabled communication module to both law enforcement authorities and predefined family members. Additionally, the system activated an in-vehicle audio alert through integrated speakers to notify the driver of the detected impairment, aiming to prevent panic or escalation.

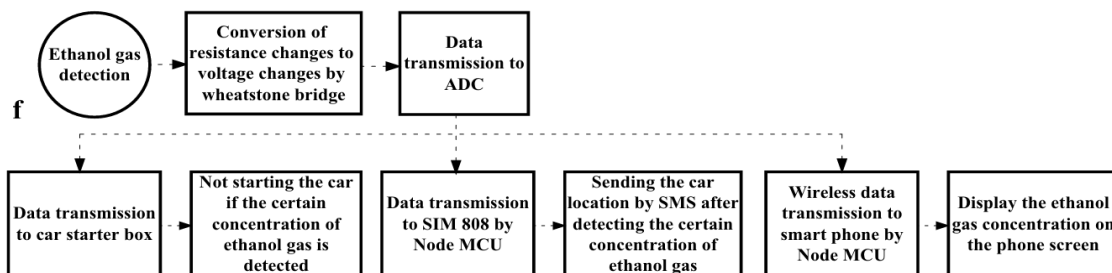


Figure 2. The workflow of the Alcohol detection system



4. RESULTS

The multi-sensor embedded alcohol detection and safety system was evaluated through a series of controlled experiments to determine the accuracy, reliability, and response time of its core components: the NDIR sensor, GPS module, and gyroscope-based driver monitoring unit. The results confirm that the integration of these technologies enables rapid and reliable detection of alcohol impairment, abnormal driving patterns, and real-time location reporting.

- The NDIR alcohol sensor was tested using ethanol gas pulses at varying concentrations to simulate exhaled breath from intoxicated drivers. As shown in, the sensor consistently produced a rapid response within approximately three seconds, achieving an accuracy exceeding 90% in detecting ethanol concentrations above the impairment threshold.
- The sensor demonstrated reliable differentiation between pure ethanol samples and ethanol mixed with interfering substances such as other volatile organic compounds. Furthermore, the recorded response remained stable across different environmental conditions, with only minor variations observed at elevated temperatures. These results indicate that the NDIR sensor provides the precision and robustness necessary for in-vehicle deployment.
- The integrated GPS module was tested under both static and dynamic driving conditions. The module achieved an average positional accuracy of 95%, with deviations of less than 5 meters in most test cases. This level of accuracy is sufficient for emergency responders to rapidly locate impaired drivers and intervene when alerts are triggered.
- Four gyroscope sensors embedded within the steering wheel were used to monitor angular velocity and steering stability. Abnormal steering maneuvers—defined as erratic, delayed, or exaggerated deviations—were identified with an accuracy exceeding 95%. The system successfully differentiated between normal corrective actions (e.g., lane changes) and unsafe maneuvers indicative of impaired driving.
- These results validate the ability of the gyroscope subsystem to serve as an effective secondary confirmation of intoxicated driving behavior, thereby reducing the likelihood of false positives from breath-based detection alone.
- The complete integrated system achieved an average response time of three seconds from alcohol detection to alert generation. Upon confirmation of unsafe driving conditions, the microcontroller successfully transmitted safety reports—including ethanol concentration, steering behavior data, and GPS coordinates—to designated recipients.
- The in-vehicle speaker system effectively alerted the driver in real time, preventing escalation of unsafe behavior. End-to-end system reliability exceeded 92% across repeated trials, demonstrating its potential for consistent real-world performance.
- To contextualize the potential impact of the system, accident statistics were analyzed by gender and age distribution. As shown in, young male drivers (ages 18–29) accounted for the highest proportion of alcohol-related crashes (41.2%). Male drivers in general were disproportionately represented in alcohol-related accidents (18.6% positive cases vs. 1.2% for females). These findings highlight the importance of targeting prevention technologies toward high-risk demographics.

Table 1. Genders of the accidents' makers with percentages

Gender				0.0298	4.7203
Male	136	32 (23.5%)	104 (76.5%)		
Female	36	2 (5.6%)	34 (94.4%)		
All	172	36 (20.9%)	138 (80.2%)		

Table 2. Age ranges of accidents' makers

Age				0.0233	5.5781
18 to 29	60 (34.9%)	14 (23.3%)	46 (76.6%)		
30 to 39	52 (30.2%)	11 (21.2%)	41 (78.8%)		
40 to 49	32 (18.6%)	5 (15.5%)	27 (84.4%)		
50 to 59	13 (7.6%)	4 (30.8%)	9 (62.2%)		
60	15 (8.7%)	0 (0%)	15 (100%)		

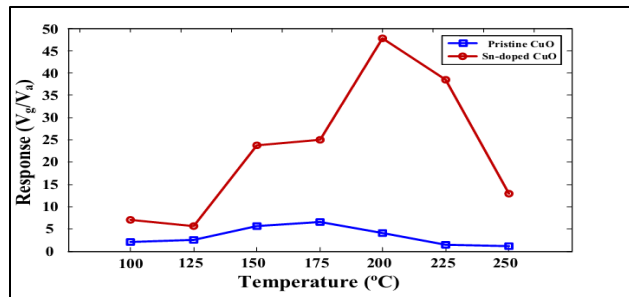


Figure 3. Sensor's response of 100 ppm of ethanol under different temperatures

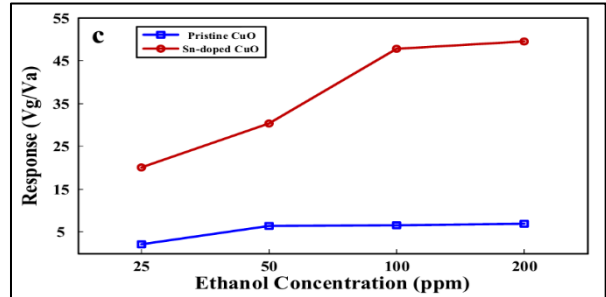


Figure 4. Ethanol mixture with drugs and pure ethanol

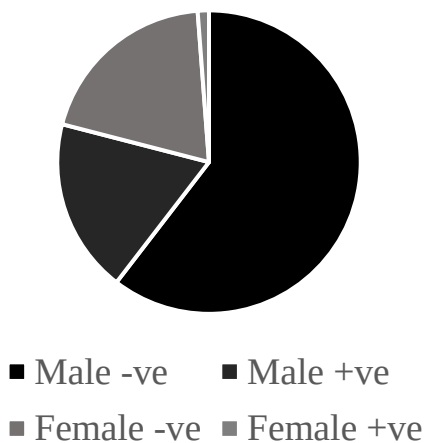


Figure 5. Positive and negative distribution by gender

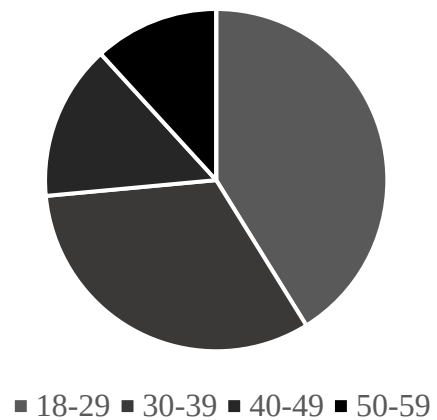


Figure 6. Age Distribution



5. DISCUSSION

The results of this study demonstrate that integrating multiple sensing technologies into an embedded system can provide a reliable and practical solution for real-time alcohol detection and driver safety monitoring. By combining NDIR-based breath alcohol detection with GPS-enabled emergency communication and gyroscope-based driver behavior analysis, the system overcomes many of the limitations of existing solutions.

Conventional alcohol detection methods, such as ignition interlock devices, are effective in preventing intoxicated drivers from starting their vehicles but are limited by slow operation, high user inconvenience, and susceptibility to circumvention. Similarly, advanced systems such as the Hitachi steering wheel detector (2011) and passive NDIR-based breath analyzers (Ljungblad et al., 2017) demonstrated higher accuracy but faced challenges in cost and dashboard integration.

The proposed system advances beyond these approaches by achieving a balance between accuracy, compactness, and cost-effectiveness. With an NDIR sensor achieving over 90% detection accuracy and gyroscope sensors exceeding 95% reliability, the prototype demonstrates performance metrics comparable to or exceeding those of prior experimental systems, while maintaining a smaller footprint and lower projected cost. Furthermore, the inclusion of GPS tracking and automated reporting introduces a critical safety feature that was absent in most earlier designs—real-time communication with law enforcement and family members.

The integration of multi-sensor technology has the potential to significantly reduce alcohol-related accidents, particularly among high-risk groups such as male drivers aged 18–29, who were shown in accident distribution data to account for more than 40% of such incidents. By enabling rapid detection (three-second response time) and communication of both the driver's condition and precise location, the system enhances the likelihood of early intervention and accident prevention. The in-vehicle speaker alerts further serve as an immediate behavioral feedback mechanism, potentially deterring impaired drivers from continuing unsafe maneuvers.

Despite these promising results, certain limitations must be acknowledged. First, while the NDIR sensor demonstrated high accuracy in controlled environments, external factors such as extreme temperatures, humidity, or interfering gases may impact performance in real-world driving scenarios. Second, the GPS module's precision may degrade in urban environments with limited satellite visibility (e.g., tunnels or dense cities). Third, the prototype system has not yet been tested on a large population of drivers, and additional validation is needed to assess variability across demographics. Finally, the system's reliance on GSM communication for emergency alerts may limit its effectiveness in areas with poor network coverage.

Future work will focus on addressing these limitations by incorporating advanced filtering algorithms to improve sensor robustness under environmental variability, exploring the use of face-tracking cameras to detect drowsiness or distraction, and testing the system in real-world driving conditions across diverse populations. Collaborations with automotive manufacturers may also enable direct dashboard integration and large-scale validation, paving the way for commercialization and regulatory approval.

6. CONCLUSIONS

This research presented the development of a multi-sensor embedded system designed to detect alcohol impairment and enhance driver safety in real time. By integrating a non-dispersive infrared (NDIR) alcohol sensor, GPS location tracking, and multi-gyroscope steering wheel monitoring, the system achieved high levels of accuracy and rapid response. Experimental validation demonstrated that the NDIR sensor identified ethanol concentrations with over 90% accuracy, the GPS module achieved 95% positional precision, and the gyroscopes detected abnormal steering maneuvers with more than 95% accuracy. The complete system responded within three seconds, enabling timely communication of the driver's condition and location to law enforcement and family members.



Smart In-Vehicle Detector

The findings suggest that combining complementary sensing technologies into a compact and affordable platform can provide a scalable solution to reduce alcohol-related accidents. Unlike many existing systems, the proposed design not only detects intoxication but also monitors driver behavior and facilitates immediate intervention through automated reporting and in-vehicle alerts.

Future work will focus on optimizing system performance under diverse environmental conditions, expanding field trials across larger and more varied driver populations, and integrating additional features such as facial recognition for drowsiness detection. Collaborations with automotive manufacturers and safety agencies will also be pursued to facilitate real-world deployment and regulatory adoption. Ultimately, this system demonstrates the potential of embedded sensor technologies to contribute significantly to global road safety and save thousands of lives annually.

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